Update 5 – Weekly Update Report

CSE 499B: Senior Design II | Section: 12 | Group: 03

Week Update Summary

Over the last week of this period, the group has achieved a great deal of advancement in numerous program modules. The Google form automation system has been integrated with a frontend user interface, which will allow interactive, step-by-step completion of the form via chat between participants and moderators. The profile checker & resume checker functionality used a fully dynamic React UI and leveraged data from Supabase for its data management. Significant improvements were made to the AI Mock Interview program with the fine-tuning of a large language model leveraging the QLoRA approach to generate quality questions and assess participant responses in a better fashion. Furthermore, additional improvements were made to both the accuracy and usability of the model by creating a working Streamlit user interface that enabled users to interact with the model and have performance metrics provided to them on a real-time basis.

Contribution

Md. Kamrul Islam (2211745642):

In the previous week, I automated the Google Form submission. This week, I have tried to implement the frontend part. After clicking the apply button, the agent sends the text "I am starting to apply for the job. Please bear with me..." first, then opens a new Chrome tab, navigates to the link, and starts filling out the Google form using Selenium. The agent will take the question from the Google form, ask the question to the user in the chat, the user will answer it, the agent will take the answer, fill up the form with the answer, and go for the next question as before. After completing the Google form, the agent will automatically submit it by clicking the submit/apply button.

Farhan Ishraque (2212002042):

This week, I have successfully finished all my tasks for this project. In this project, my role was to complete the resumechecker part and the profile part with signup and database. I worked in the Profile part, where I used the React component (ProfileSection) to complete the user profile management UI connected to a backend (Supabase via profileservice). It allows users to view and edit personal info, contact details, work experience, and education using editable forms and tabs. It includes file upload features for profile pictures (avatar) and CV/resume with preview, download, and note functionality. I used Supabase for database management. It fetches and synchronizes user data (profile, work, education, CV metadata) from the backend when the component loads. It supports dynamic addition, editing, and deletion of work and education entries with real-time UI updates.

Iram Shehzad (2131614642):

This week, I used QLoRA on Google Colab's T4 GPU to fine-tune a Qwen2.5-7B-Instruct large language model for the AI Mock Interview module of JobCore. This model created interview questions that were specific to each role and graded user answers based on how clear, correct, relevant, and deep they were. I made a custom dataset with 341 examples that included 10 job roles, 5 topic categories, and 3 difficulty levels. Then I trained the model for 3 epochs with only 0.28% of the parameters being updated. The fine-tuned model works with a FastAPI backend for the group's React frontend and a Streamlit demo for use on its own. Some of the main problems were limited GPU memory (fixed by gradient checkpointing and a smaller batch size), unbalanced datasets (fixed by generating bad or average answers), and connecting the Colab-hosted model to the local app

<u>Atikul Islam Nahid (2211978042):</u>

This week, I focused on improving my mock interview project. First, I worked on increasing the accuracy of the AI model so it can give better and more correct answers. I made some adjustments and improvements in the training process, which helped the model perform better than before.After that, I connected the AI model with a frontend so users can interact with it easily. For this, I used Streamlit, which helped me build a simple and user-friendly interface.Now, the system runs successfully. Users can give interview questions from the front end and give answers in real time and get a summary and final score of his overall performance. Moreover,the part of my project is now more accurate and fully functional with a working user interface.

Completed Task:

- A complete Profile page with Supabase database with signup.
- Various model training with the dataset.
- Read, write, and automatic Google Form submission.
- Complete Merge of all the features.

Work in Progress

- Mock interview voice generation.
- Add a mock interview custom model to the main frontend.
- Upload files to a Google Form and submit it automatically using the website.

Next Target

- Integrate the custom model into the mock interview feature.
- Fully read and write on a Google Form and submit it automatically using the website.
- To make fully workable prototype.

Challenges

- Bypassing Google Chrome’s automation blocking.
- Costly existing tools or websites in the current market.